

Milind Bansal

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EDUCATION

Your University Name

Bachelor of Technology in Computer Science / AI & Machine Learning

City, Country

Expected Month YYYY

- GPA: X.X/10 • Relevant Coursework: Machine Learning, Deep Learning, NLP, Statistics, Linear Algebra, Algorithms

TECHNICAL SKILLS

Machine Learning: PyTorch, Hugging Face Transformers, scikit-learn, DistilBERT, BERT, Fine-tuning, Transfer Learning

NLP: Tokenization, Sequence Classification, Token-Level Interpretability, Bias & Fairness Evaluation

LLM & GenAI: Groq API, LLaMA 3.3 70B, OpenAI SDK (compatible), Structured Outputs, Prompt Engineering

MLOps: Docker, Docker Compose, FastAPI, Uvicorn, Model Serving, Containerized ML Pipelines

Evaluation: F1-Score, ROC-AUC, Precision/Recall, Confusion Matrix, Bias Testing

Backend: Python (FastAPI), Node.js (Express, Next.js), TypeScript, PostgreSQL, Prisma, Redis

Tools: Git, Jupyter Notebooks, Weights & Biases (WandB), Pandas, NumPy

PROJECTS

TruthLens | *PyTorch, DistilBERT, Hugging Face, FastAPI, React, Docker Compose, PostgreSQL* 🔄 2024 – 2025

- [S] Existing AI-content detectors operated as opaque black boxes, providing predictions without explanations, eroding user trust; [T] built an end-to-end ML platform with interpretability as a first-class feature; [A] fine-tuned **distilbert-base-uncased** on a balanced 10,000-sample dataset (GPT-4, Claude, human text) using PyTorch and Hugging Face, implementing token-level SHAP-style confidence heatmaps to surface which tokens drove the prediction; [R] achieved **91.3% F1-score** and **93.1% accuracy**, with **<300ms inference latency on CPU**—no GPU required.
- Designed and served the model via a **FastAPI** inference service (Uvicorn, async endpoints) containerized in Docker Compose alongside PostgreSQL and a React dashboard, enabling one-command zero-config deployment.
- Implemented a rigorous **bias & fairness evaluation** across text genres (news articles, essays, casual writing) using ROC-AUC and per-class confusion matrices, ensuring the model did not overfit to stylistic artifacts of a single genre.
- Chose DistilBERT over BERT to gain **60% speed improvement** and **40% model size reduction** while retaining 97% of accuracy—a deliberate engineering tradeoff enabling CPU-only production deployment.

FeedbackOS | *Groq API (LLaMA 3.3 70B), BullMQ, Next.js 16, PostgreSQL, Redis, TypeScript* 🔄 🌐 Live 2025

- [S] Manual PM feedback classification consumed 5–10 hours/week per team; [T] automate multi-source feedback ingestion and classification using production-grade LLM pipelines; [A] architected an agentic classification pipeline using Groq's **LLaMA 3.3 70B** via structured outputs (OpenAI-compatible SDK), returning JSON schema-constrained fields (theme, sentiment, product_area, confidence, requires_human_review), decoupled from HTTP via BullMQ async queues; [R] achieved **~1.2s end-to-end latency** per item with HTTP response times remaining unaffected.
- Secured all third-party integration credentials (Zendesk API tokens) with **AES-256-GCM encryption at rest**, demonstrating production security practices for AI integrations handling sensitive data.
- Implemented structured **LLM output validation** using Zod schemas on the parsed JSON response, ensuring type-safe persistence even when the LLM returns unexpected formats.

MedMarket India – AI/Analytics Features | *Node.js, TypeScript, Gemini API, Groq LLaMA 3.3* 🔄 2024 – 2025

- Integrated a **dual-provider AI setup** (Gemini primary, Groq LLaMA 3.3 70B fallback) for automated pharmacy document verification and product insights, providing production-grade AI reliability with zero single-point-of-failure.
- Built a **pharmacy analytics engine** computing 14-day revenue trends, hourly order heatmaps, fulfillment/rejection rates, repeat-customer ratios, and dead-stock value-at-risk—surfacing actionable ML-adjacent insights from structured database aggregations.

RouteLens | *Next.js 14 (Static Export), TypeScript, OpenAPI/Swagger Parsing* 🔄 🌐 Live 2024

- Built a client-side YAML/JSON parser with full **\$ref** resolution for OpenAPI 2.0/3.x specs, demonstrating applied understanding of structured data schema interpretation—a skill directly applicable to ML data pipeline design.

CERTIFICATIONS & ACHIEVEMENTS

Digital Heroes Programme – Certificate of Completion (2024)

TruthLens: Fine-tuned DistilBERT achieving 91.3% F1 on AI-generated content detection with bias-fairness evaluation

FeedbackOS: Production LLM integration (Groq LLaMA 3.3 70B) with structured output validation and async processing